

9th Mathematics - English MCQs

Clean single-column pattern | Answer key at the end

Chapter 1

1. The order of matrix $\begin{bmatrix} 2 & 1 \end{bmatrix}$ is ...

- A. 2-by-1
- B. 1-by-2
- C. 1-by-1
- D. 2-by-2

2. $\begin{bmatrix} 2 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$ is called Matrix.

- A. zero
- B. unit
- C. scalar
- D. singular

3. Which is order of a square matrix?

- A. 2-by-2
- B. 1-by-2
- C. 2-by-1
- D. 3-by-2

4. Which is order of a rectangular matrix?

- A. 2-by-2
- B. 4-by-4
- C. 2-by-1
- D. 3-by-3

5. Order of transpose of $\begin{bmatrix} 2 & 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ is ...

- A. 3-by-2
- B. 2-by-3
- C. 1-by-3
- D. 3-by-1

6. Adjoint of $\begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ is

- A. $\begin{bmatrix} 2 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- B. $\begin{bmatrix} 2 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- C. $\begin{bmatrix} 2 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- D. $\begin{bmatrix} 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$

7. If $\begin{bmatrix} 2 & 6 & 0 & 3 \end{bmatrix} x = \begin{bmatrix} 9 & -6 & 6 & -9 \end{bmatrix}$, then x is equal to

- A. 9
- B. -6
- C. 6
- D. -9

8. Product of $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ is

- A. $\begin{bmatrix} 2x & y \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- B. $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2y & -1 \end{bmatrix}$
- C. $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2x & y \end{bmatrix}$
- D. $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2y & -1 \end{bmatrix}$

9. $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -1 & -2 & 0 \end{bmatrix}$ If $X + \begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$ then X is equal to

- A. $\begin{bmatrix} 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- B. $\begin{bmatrix} 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- C. $\begin{bmatrix} 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$
- D. $\begin{bmatrix} 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$

10. The idea of a matrices was given by:___

- A. Arthur Cayley
- B. Leonard Euler
- C. Henry Briggs
- D. John Napier

11. If $2A + 3B - C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then $-A =$ _____

- A. $2A + 3B - C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- B. $2A + 3B - C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- C. $2A + 3B - C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- D. $2A + 3B - C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

12. A square matrix is symmetric if ____

- A. $A^T = A$
- B. $A^{-1} = A$
- C. $(A^T)^T = -A^T$
- D. $A^T = -A$

13. A square matrix is skew-symmetric if:

- A. $A^T = -A$
- B. $A^{-1} = -A$
- C.
- D. $A^T = A$

14. A square matrix A is called singular if

- A. $|A| \neq 0$
- B. $|A| = 0$
- C. $A = 0$
- D. $A^T = 0$

15. A square matrix A is called non-singular if:

- A. $|A| = 0$
- B. $A = 0$
- C. $|A| \neq 0$
- D. $A^T = 0$

16. $(AB)^{-1} =$ _____

- A. $A^{-1} B^{-1}$
- B. $B^{-1} A^{-1}$
- C. BA
- D. AB

17. Additive inverse of $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$ is _____

- A. $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$
- B. $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$
- C. $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$
- D. $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$

18. If A is a matrix then its transpose is denoted by:

- A. A^{-1}
- B. A^T
- C. $-A$
- D. $(A^T)^T$

19. Which of the following is singular matrix?

- A. $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 3 & 4 \end{bmatrix}$
- B. $\begin{bmatrix} 1 & 3 & 4 \\ 2 & 3 & 4 \end{bmatrix}$
- C. $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 2 & 3 & 4 & 5 \end{bmatrix}$
- D. $\begin{bmatrix} 1 & 3 & 6 \\ 2 & 3 & 4 \end{bmatrix}$

20. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then the det. A is:

- A. $ad - bc$
- B. $bc - ad$
- C. $ad + bc$
- D. $bc + ad$

Chapter 2

1. $(-1)^{327}x = \underline{\hspace{2cm}}$

- A. $39x$
- B. $39x$
- C. $38x$
- D. $38x$

2. Write 7^x in exponential form

- A. x
- B. x^7
- C. $17x$
- D. $7x$

3. Write 3^4 with radical sign....

- A. 3^4
- B. 3^4
- C. 2^3^4
- D. 6^4

4. In 3^35 the radicand is

- A. 3
- B. 1^3
- C. 35
- D. None of these

5. $-\sqrt[4]{125} = \underline{\hspace{2cm}} 16$

- A. 5^4
- B. 4^5
- C. $5^4 -$
- D. $4^5 -$

6. The conjugate of $5 + 4i$ is $\underline{\hspace{2cm}}$

- A. $-5 + 4i$
- B. $-5 - 4i$
- C. $5 - 4i$
- D. $5 + 4i$

7. The value of i^9 is $\underline{\hspace{2cm}}$

- A. 1
- B. -1
- C. i
- D. $-i$

8. Every real number is $\underline{\hspace{2cm}}$

- A. A positive integer
- B. A rational number
- C. A negative integer
- D. A complex number

9. Real part of $2ab(i^2 + i)$ is $\underline{\hspace{2cm}}$

- A. $2ab$
- B. $-2ab$
- C. $2abi$
- D. $-2abi$

10. Imaginary part of $-i(3i + 2)$ is_

- A. -2
- B. 2
- C. 3
- D. -3

11. Which of the following sets have the closure property w.r.t. addition_____

- A. $\{0\}$
- B. $\{0, -1\}$
- C. $\{0, 1\}$
- D. $1, 2, 2 \times 2, 2 \times 2 \times 2, 2 \times 2 \times 2 \times 2$

12. Name the property of real numbers used in $5 \times 1 = 5$ - - $5 \times 1 = 5$

- A. Additive identity
- B. Additive Inverse
- C. Multiplicative identity
- D. Multiplicative Inverse

13. If x, y, z in $\mathbb{R}$ $z < 0$ then $x < y \Rightarrow$

- A. $xz < yz$
- B. $xz > yz$
- C. $xz = yz$
- D. none of these

14. If a, b in $\mathbb{R}$ then only one of $a = b$ or $a < b$ or $a > b$ holds is called...

- A. Trichotomy property
- B. Transitive property
- C. Additive property
- D. Multiplicative property

15. A non-terminating, non-recurring decimal represents:

- A. A natural number
- B. A rational number
- C. An irrational number
- D. A prime number

16. The union of the set of rational numbers and irrational numbers is known as set of ____

- A. Rational number
- B. Irrational
- C. Real number
- D. Whole number

17.

18. $nab =$ ____

- A. $nna b$
- B. $a b$
- C. $na b$
- D. $na b$

19. $-5 \times 8 =$ - - - - -

- A. $15 \times (8) -$
- B. $(-8)5$
- C. (-8)
- D. $15 \times (8)$

20. The value of i^{10} is:

- A. -1
- B. 1
- C. -i
- D. i

21. The conjugate of $2 + 3i$ is ____

- A. $2 - 3i$
- B. $-2 - 3i$
- C. $-2 + 3i$
- D. $2 + 3i$

22. Real part of $(2 - 2i)^2 + 2$ is:

- A. -1
- B. $2\sqrt{2}$
- C. 1
- D. $2\sqrt{2}$

23. Imaginary part of $(2 - i)^2 + 2$ is

- A. -1
- B. $2\sqrt{2}$
- C. 1
- D. $2\sqrt{2}$

24. $\sqrt{2}$ is a/an.....number

- A. irrational
- B. rational
- C. natural
- D. whole

25. The value of i (iota) is ____

- A. 1
- B. -1
- C. +1
- D. $(-1)^2$

26. In $-2 + 3i$, 3 is called ____

- A. imaginary part
- B. real part
- C. negative part
- D. complex number

27. The set of natural numbers is.....

- A. $\{0, 1, 2, 3, \dots\}$
- B. $\{2, 4, 6, \dots\}$
- C. $\{1, 2, 3, \dots\}$
- D. $\{2, 3, 5, 7, \dots\}$

28. π , e , $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{5}$ are called...

- A. irrational numbers
- B. rational number
- C. natural numbers
- D. real number

29. If $x + iy = 4 + 3i$, then x and y are

- A. $x = 4, y = 3$
- B. $x = 3, y = 4$
- C. $x = 4, y = 3$
- D. $x = 3, y = 4$

30. $\sqrt{3}$ form of 0.3 is ____.

- A. $\frac{3}{10}$
- B. $\frac{1}{3}$
- C. 0.33
- D. $\frac{10}{3}$

Chapter 3

1. If $ax = n$, then _____

- A. $a = x \log n$
- B. $x = \log n a$
- C. $x = a \log n$
- D. $a = n \log x$

2. The relation of $y = \log_z x$ implies

- A. $yx z =$
- B. $yz x =$
- C. $zx y =$
- D. $zy x =$

3. The logarithm of unity to any base is

- A. 1
- B. 10
- C. e
- D. 0

4. The logarithm of any number to itself as base is _____

- A. 1
- B. 0
- C. -1
- D. 10

5. $\log e =$ _____ where $e \sim 2.718$

- A. 0
- B. 0.4343
- C. $\frac{1}{e}$
- D. 1

6. The value of $\log_p \frac{p}{q}$ is _____

- A. $\log p - \log q$
- B. $\log p \log q$
- C. $\log p + \log q$
- D. $\log q - \log p$

7. $\log p - \log q$ is same as:

- A. $q \log p$
- B. $(\) \log p q -$
- C. $\log p \log q$
- D. $p \log q$

8. $\log n^m$ can be written as

- A. $(\log m)^n$
- B. $m \log n$
- C. $n \log m$
- D. $\log (m n)$

9. $b^c \log a \times \log b$ can be written as _____

- A. $c \log a$
- B. $a \log c$
- C. $a \log b$
- D. $b \log c$

10. $\log_y x$ will be equal to _____

- A. $z y \log x \log z$
- B. $x y \log z \log z$
- C. $z z \log x \log y$
- D. $y z z \log \log x$

11. For common logarithm, the base is_

- A. 2
- B. 10
- C. e
- D. 1

12. For natural logarithm, the base is__

- A. 10
- B. e
- C. 2
- D. 1

13. The integral part of the common logarithm of a number is called the_

- A. Characteristic
- B. Mantissa
- C. Logarithm
- D. None

14. The decimal part of the common logarithm of a number is called the ____:

- A. Characteristic
- B. Mantissa
- C. Logarithm
- D. None

15. If $x = \log y$, then y is called the ____ of x.

- A. Antilogarithm
- B. Logarithm
- C. Characteristic
- D. None

16. 30600 in scientific notation is __

- A. 3.06×10^4
- B. 3.006×10^4
- C. 30.6×10^4
- D. 306×10^4

17. 6.35×10^6 in ordinary notation is__

- A. 6350000
- B. 635000
- C. 6350
- D. 63500

18. A number written in the form $a \times 10^n$, where $1 \leq a < 10$ and n is an integer is called ____

- A. Scientific notation
- B. Ordinary notation
- C. Logarithm notation
- D. None

19. Common logarithm is also known as _____ logarithm.

- A. natural
- B. simple
- C. scientific
- D. decadic

20. $\log a + \log m + \log n$ is same as:

- A. $\log a m n +$
- B. $\log a m n \times$
- C. $\log \log a a m n \times$
- D. $\log a m n$

21. John Napier prepared the logarithms tables to the base ____.

- A. 0
- B. 1
- C. 10
- D. e

22. $3^2 \log$ in common logarithm is written as ____.

- A. $\log 3 \log 2$
- B. $\log 2 \log 3$
- C. $\log 3^2$
- D. $3 \log 2$

23. $e \log 10 =$ ____

- A. 2.3026
- B. 0.4343
- C. $10 e$
- D. 10

24. If $x^2 \log = 5$ then x is:

- A. 25
- B. 32
- C. 10
- D. $2 \times$

Chapter 4

1. $4x + 3y - 2$ is an algebraic ____

- A. Expression
- B. Sentence
- C. Equation
- D. In equation

2. The degree of polynomial $4x^4 + 2x^2y$ is ____

- A. 1
- B. 2
- C. 3
- D. 4

3. $a^3 + b^3$ is equal to ____

- A. $(a-b)(a^2+ab+b^2)$
- B. $(a+b)(a^2-ab+b^2)$
- C. $(a-b)(a^2-ab+b^2)$
- D. $(a-b)(a^2+ab-b^2)$

4. $(-)(-)^3 2^3 2^2 + -$ is equal to: ____

- A. 7
- B. -7
- C. -1
- D. 1

5. Conjugate of Surd $a + b\sqrt{}$ is _

- A. $a + b\sqrt{}$
- B. $a - b\sqrt{}$
- C. $a + b\sqrt{}$
- D. $a - b\sqrt{}$

6. $(a+b)(a-b)(a^2+b^2)(a^4+b^4)$ is equal to

- A. $a^8 - b^8$
- B. $a^8 + b^8$
- C. $a^8 - b^8$
- D. $a^8 + b^8$

7. $2a^2b - ab^2 +$ is equal to:

- A. $(a-b)^2$
- B. $(a+b)^2$
- C. $a+b$
- D. $a-b$

8. $()a^2b + ()ab^2 -$ is equal to: __

- A. $a^2 + b^2$
- B. $a^2 - b^2$
- C. $a - b$
- D. $a + b$

9. The degree of the polynomial $x^2y^2 + 3xy + y^3$ is __

- A. 4
- B. 5
- C. 6
- D. 2

10. $2x^4 - =$

- A. $(x-2)(x+2)$
- B. $(x-2)(x-2)$
- C. $(x+2)(x+2)$
- D. $(x-2)^2$

11. $3x^2 - 11x + 12 = (.....)$

- A. $2x^2 - 11x + 12$
- B. $2x^2 - 11x + 12$
- C. $2x^2 - 11x + 12$
- D. $2x^2 - 11x + 12$

12. $1 \frac{2}{3} = -$

- A. $2 \frac{3}{4}$
- B. $2 \frac{3}{4}$
- D. $2 \frac{3}{4} - +$

13. $(a+b)^2 - (a-b)^2 =$ _____

- A. $2(a^2 + b^2)$
- B. $4ab$
- C. $2ab$
- D. $3ab$

14. A surd which contains a single term is called ____ surd.

- A. Monomial
- B. Binomial
- C. Trinomial
- D. Conjugate

15. What is the leading coefficient of polynomial $2x^3 + 8x + 5$?

- A. 2
- B. 3
- C. 5
- D. 8

16. A surd which contains two terms is called ____ surd.

- A. Monomial
- B. Binomial
- C. Trinomial
- D. Conjugate

17. Which of the following is polynomial?

- A. $2x^3 +$
- B. $2x^4 - 3x^2 -$
- C. $2x^3 - 2x^2 +$
- D. $2x^3 - 2x^2 +$

18. $() () 3 + 3 3 - 3 = \underline{\hspace{2cm}}$

- A. 12
- B. 9
- C. 6
- D. 3

19. Which of the following is not surd?

- A. 2
- B. 3
- C. $2\sqrt{5}$
- D. π

20. In the polynomial with the variable x, all the powers of x are----- integers.

- A. non-negative
- B. negative
- C. non-positive
- D. none of these

21. Polynomial means an expression with:

- A. one term
- B. two terms
- C. three terms
- D. many term

correct answer.

Chapter 5 - Factorization

1. The factor of $x^2 - 5x + 6$ are: __

- A. $x + 1, x - 6$
- B. $x - 2, x - 3$
- C. $x + 6, x - 1$
- D. $x + 2, x + 3$

2. Factors of $8x^3 + 27y^3$ are: __

- A. $(2x+3y)(4x^2-9y^2)$
- B. $(2x-3y)(4x^2 - 9y^2)$
- C. $(2x + 3y)(4x^2 - 6xy + 9y^2)$
- D. $(2x-3y)(4x^2 + 6xy + 9y^2)$

3. Factors of $3x^2 - x - 2$ are:

- A. $(x+1)(3x-2)$
- B. $(x+1)(3x+2)$
- C. $(x-1)(3x-2)$
- D. $(x-1)(3x+2)$

4. Factors of $a^4 - 4b^4$ are: __

- A. $(a-b)(a+b)(a^2+4b^2)$
- B. $(a^2-2b^2)(a^2 + 2b^2)$
- C. $(a-b)(a+b)(a^2-4b^2)$
- D. $(a-2b)(a^2+ 2b^2)$

5. What will be added to complete the square of $9a^2 - 12ab$? __

- A. $-16b^2$
- B. $16b^2$
- C. $4b^2$
- D. $-4b^2$

6. Find m so that $x^2 + 4x + m$ is a complete square:

- A. 8
- B. -8
- C. 4
- D. 16

7. Factors of $5x^2 - 17xy - 12y^2$ are__

- A. $(x+4y) (5x+3y)$
- B. $(x-4y) (5x - 3y)$
- C. $(x-4y) (5x + 3y)$
- D. $(5x - 4y) (x +3y)$

8. Factors of $3^3 127x^3 -$ are__

- A. $2^2 11 3x 9x 3 \times \times \square \square \square - + + \square \square \square \square \square \square$
- B. $2^2 11 1 3x 9x 3 \times \times \square \square \square + + + \square \square \square \square \square \square$
- C. $2^2 11 1 3x 9x 3 \times \times \square \square \square - - + \square \square \square \square \square \square$
- D. $2^2 11 1 3x 9x 3 \times \times \square \square \square + - + \square \square \square \square \square \square$

9. If $x-2$ is a factor of $p(x) = x^2+2kx+8$, then $k =$ __

- A. -3
- B. 3
- C. 4
- D. 5

10. $4a^2+4ab+(.....)$ is a complete square

- A. b^2
- B. $2b$
- C. a^2
- D. $4b^2$

11. $-2^2 2^2 x y^2 + = y x$

- A. $2 x y y x \square \square - \square \square \square$
- B. $2 x y y x \square \square + \square \square \square$
- C. $3 x y y x \square \square - \square \square \square$
- D. $3 x y y x \square \square + \square \square \square$

12. $(x+y) (x^2 - xy + y^2) =$ __

- A. $x^3- y^3$
- B. $x^3 + y^3$
- C. $(x+y)^3$
- D. $(x - y)^3$

13. Factors of $x^4 - 16$ is __

- A. $(x-2)^2$
- B. $(x-2) (x+2) (x^2+4)$
- C. $(x-2) (x+2)$
- D. $(x+2)^2$

14. Factors of $3x - 3a + xy - ay$.

- A. $(3+y) (x-a)$
- B. $(3-y) (x+a)$
- C. $(3-y) (x-a)$
- D. $(3+y) (x+a)$

15. Factors of $pqr + qr^2 -pr^2 - r^3$ is:

- A. $r(p+r) (q-r)$
- B. $r(p-r) (q + r)$
- C. $r(p-r) (q-r)$
- D. $r(p+r) (q+r)$

16. What is the value of $()^4 3 p x = 6x + 2x - x + 2$ at $x = 0$?

- A. 9
- B. 8
- C. 2
- D. 7

17. $2x^2 + 5x + 6 =$

- A. $(x+2)(2x+3)$
- B. $(x+2)(3x+2)$
- C. $(x+3)(2x+2)$
- D. $(x+2)(3x+3)$

18. $24a^2 - 16 =$

- A. $2(8a^2 - 8a + 2)$
- B. $4(2a^2 - 2a + 2)$
- C. $2(4a^2 - 2a + 2)$
- D. $2(4a^2 - 2a - 2)$

19. How many factors of a cubic expression are there?

- A. zero
 - B. 1
 - C. 2
 - D. 3
- $20. (x - y)(x^2 + xy + y^2) =$ ____

Chapter 6

1. H.C.F of $p^3q - pq^3$ and $p^5q^2 - p^2q^5$ is _

- A. $pq(p^2 - q^2)$
- B. $pq(p - q)$
- C. $p^2q^2(p - q)$
- D. $pq(p^3 - q^3)$

2. H.C.F. of $5x^2y^2$ and $20x^3y^3$ is: ____

- A. $5x^2y^2$
- B. $20x^3y^3$
- C. $100x^5y^5$
- D. $5xy$

3. H.C.F of $x^2 - 2$ and $x^2 + x - 6$ is _

- A. $x^2 + x - 6$
- B. $x + 2$
- C. $x - 2$
- D. $x + 3$

4. H.C.F of $a^3 + b^3$ and $a^2 - ab + b^2$ is

- A. $a + b$
- B. $a^2 - ab + b^2$
- C. $(a - b)^2$
- D. $a^2 + b^2$

5. H.C.F of $x^2 - 5x + 6$ and $x^2 - x - 6$ is __:

- A. $x - 3$
- B. $x + 2$
- C. $x^2 - 4$
- D. $x - 2$

6. H.C.F of $a^2 - b^2$ and $a^3 - b^3$ is ____

- A. $a - b$
- B. $a + b$
- C. $a^2 + ab + b^2$
- D. $a^2 - ab + b^2$

7. H.C.F of $x^2 + 3x + 2$, $x^2 + 4x + 3$, $x^2 + 5x + 4$ is:

- A. $x + 1$
- B. $(x+1)(x+2)$
- C. $(x+3)$
- D. $(x+4)(x+1)$

8. L.C.M of $15x^2$, $45xy$ and $30xyz$ is:

- A. $90xyz$
- B. $90x^2yz$
- C. $15xyz$
- D. $15x^2yz$

9. L.C.M of a^2+b^2 and a^4-b^4 is: __

- A. $a^2 + b^2$
- B. $a^2 - b^2$
- C. $a^4 - b^4$
- D. $a - b$

10. The product of two algebraic expression is equal to the __ of their H.C.F and L.C.M.

- A. Sum
- B. Difference
- C. Product
- D. Quotient

11. Simplify $2^2 a^1 + 3a - b$ $9a - b$

- A. $2^2 4a^9a b -$
- B. $2^2 4a b^9a b - -$
- C. $2^2 4a b^9a b + -$
- D. $2^2 b^9a b -$

12. Simplify $2^2 a + 5a - 14a + 3 \times a - 2a - 3a - 18 =$

- A. $a^7 a^6 + -$
- B. $a^7 a^2 + -$
- C. $a^3 a^6 + -$
- D. $a^2 a^3 - +$

13. Simplify $3^3 4^4 a - b a - b$ $2^2 2^2 a + ab + b a + b =$

- A. $1 a b +$
- B. $1 a b -$
- C. $2^2 a b a b - +$
- D. $2^2 a b a b + +$

14. Simplify: $2x + y$ $x - 1 \div 1 - x + y$ $x + y$

- A. $x \times y +$
- B. $y \times y +$
- C. $y \times$
- D. $x \times y$

15. The square root of $a^2 - 2a + 1$ is _

- A. $\sqrt{a+1}$
- B. $\sqrt{a-1}$
- C. $a-1$
- D. $a+1$

16. What should be added to complete the square of $x^4 + 64$?

- A. $8x^2$
- B. $-8x^2$
- C. $16x^2$
- D. $4x^2$

17. The square root of $4^4 1 x + + 2 x$ is __

- A. $1 \times x \sqrt{\quad} + \sqrt{\quad}$
- B. $2^2 1 \times x \sqrt{\quad} + \sqrt{\quad}$
- C. $1 \times x \sqrt{\quad} - \sqrt{\quad}$
- D. $2^2 1 \times x \sqrt{\quad} - \sqrt{\quad}$

18. The square root of $4x^2-12x+9$ is:

- A. $\sqrt{2x - 3}$
- B. $\sqrt{2x + 3}$
- C. $(2x + 3)^2$
- D. $(2x - 3)^2$

19. L.C.M = ____

- A. $p(x) q(x)$ H.C.F x
- B. $p(x) q(x)$ L.C.M x
- C. $p(x) q(x)$ H.C.F x
- D. $q(x) p(x)$ H.C.F x

20. H.C.F. = ____

- A. $p(x) q(x)$ L.C.M x
- B. $p(x) q(x)$ H.C.F x
- C. $p(x) q(x)$ L.C.M x
- D. L.C.M $p(x) q(x) x$

21. L.C.M $\times$ H.C.F =.....

- A. $p(x) q(x) x$
- B. $p(x)$ H.C.F x
- C. $q(x)$ L.C.M x
- D. None

22. Any unknown expression may be found if ____ of them are known by using the relation L.C.M $\times$ H.C.F = $p(x) \times q(x)$

- A. Two
- B. Three
- C. Four
- D. None

23. The H.C.F of $2x^2 - 4$, $x^2 + 4x + 4$ and $2x^2 + x - 6$ is: L.C.M $\times$ H.C.F = $p(x) \times q(x)$

- A. $2x -$
- B. $2x +$
- C. $(x - 2)(x + 3)$
- D. $(x - 2)(x + 3)(x - 2)(x + 3)(x - 2)(x + 3)$

24. $2a^2 + b^2 - ab \div a - b$ a - $2ab + b$

- A. $a b$
- B. $b a$
- C. $1 a$
- D. a

25. If $a + b$ and $a - b$, then $1/a$ is:

- A. $a b a b - +$
- B. $a b a b + -$
- C. $a b a b - -$
- D. $a b a b + +$

26. How many methods are used to find H.C.F of given expressions?

- A. one
- B. two
- C. three
- D. four

27. How many methods are used to find square root of given expression?

- A. one
- B. two
- C. three
- D. four

28. If $() () () q x . q x = p x$, then $() q x$ is called ____ of $() p x$.

- A. square
- B. square root
- C. L.C.M.
- D. H.C.F.

Chapter 7

1. Which of the following is the solution of the inequality $3 - 4x \leq 11$?

- A. $x \geq -8$
- B. $x \geq -2$
- C. $x \geq 14$
- D. None of these

2. A statement involving any of the symbols $<$, $>$ or $\leq$ or $\geq$ is called:

- A. Equation
- B. Identity
- C. Inequality
- D. Linear equation

3. $x = \underline{\hspace{1cm}}$ is a solution of the inequality $-2 < x < 3$

- A. -5
- B. 3
- C. 0
- D. 5

4. If x is no larger than 10, then:

- A. $x \geq 8$
- B. $x \leq 10$
- C. $x < 10$
- D. $x > 10$

5. If the capacity c of an elevator is at most 1600 pounds, then_

- A. $c < 1600$
- B. $c \geq 1600$
- C. $c \leq 1600$
- D. $c > 1600$

6. $x=0$ is a solution of the inequality:

- A. $x > 0$
- B. $3x + 5 < 0$
- C. $x + 2 < 0$
- D. $x - 2 < 0$

7. The linear equation in one variable x is:

- A. $ax + b = 0$
- B. $ax^2 + bx + c = 0$
- C. $ax + by + c = 0$
- D. $ax^2 + by^2 + c = 0$

8. An inconsistent equation is that whose solution set is:

- A. Empty
- B. Not empty
- C. Zero
- D. Positive

9. $x = a$ is equivalent to:

- A. $x = a$ or $x = -a$
- B. $1/x$ or $x/a = 1$
- C. $1/x = a$ or $x/a = 1$
- D. None of these

10. A linear inequality in one variable x is:

- A. $ax + b > 0, a \neq 0$
- B. $ax^2 + bx + c < 0, a \neq 0$
- C. $ax + by + c > 0, a \neq 0$
- D. $ax^2 + by^2 + c < 0, a \neq 0$

11. Law of Trichotomy is ... (a,b $\in$ R) in

- A. $a < b$ or $a = b$ or $a > b$
- B. $a < b$ or $a = b$
- C. $a < b$ or $a > b$
- D. None of these

12. Transitive law is ____

- A. $a < b$ and $b < c$, then $a < c$
- B. $a > b$ and $b < c$, then $a > c$
- C. $a > b$ and $b < c$, then $a = c$
- D. None of these

13. If $a > b, c > 0$ then:

- A. $ac < bc$
- B. $ac > bc$
- C. $ac = bc$
- D. $ac \leq bc$

14. If $a > b, c > 0$ then:

- A. $abc > c$
- B. $abc < c$
- C. $abc = c$
- D. $abc \neq c$

15. If $a > b, c < 0$, then:

- A. $abc < c$
- B. $abc > c$
- C. $abc = c$
- D. $abc \leq c$

16. If a, b $\in$ R then: $b \neq 0$

- A. $aab =$
- B. $aab =$
- C. $aab + = +$
- D. $aab - = -$

17. When the variable in an equation occurs under a radical, the equation is called a ____ equation.

- A. Radical
- B. Absolute value
- C. Linear
- D. None of these

18. $x = 0$ has only ____ solution.

- A. one
- B. two
- C. three
- D. none of these

19. The equation $x = 2$ is equivalent to:

- A. x^2 or $x^2 = -$
- B. $x = -2$ or $x = -2$
- C. $x = 2$ or $x = 1^2$
- D. $x = 2$ or $x = 1^2 -$

20. An __ is equation that is satisfied by every number for which both sides are defined:

- A. Identity
- B. Conditional
- C. Inconsistent

21. An __ equation is an equation whose solution set is the empty set:

- A. Identity
- B. Conditional
- C. Inconsistent
- D. None

22. A _ equation is an equation that is satisfied by atleast one number but is not an identity:

- A. Identity
- B. Conditional
- C. Inconsistent
- D. None

23. $x + 4 = 4 + x$ is _ equation:

- A. Identity
- B. Conditional
- C. Inconsistent
- D. None

24. $2x + 1 = 9$ is ___ equation:

- A. Identity
- B. Conditional
- C. Inconsistent
- D. None

25. $x = x + 5$ is ___ equation:

- A. Identity
- B. Conditional
- C. Inconsistent
- D. None

26. Equations having exactly the same solution are called ___ equations.

- A. equivalent
- B. Linear
- C. Inconsistent

27. A solution that does not satisfy the original equation is called ____ solution:

- A. Extraneous
- B. Root
- C. General
- D. Proper

Chapter 8

1. If $(x-1, y+1) = (0, 0)$, then (x, y) is:

- A. (1, -1)
- B. (-1, 1)
- C. (1, 1)
- D. (-1, -1)

2. If $(x, 0) = (0, y)$, then (x, y) is:

- A. (0, 1)
- B. (1, 0)
- C. (0, 0)
- D. (1, 1)

3. Point (2 -3) lies in quadrant:

- A. I
- B. II
- C. III
- D. IV

4. Point (-3, -3) lies in quadrant:

- A. I
- B. II
- C. III
- D. IV

5. If $y = 2x + 1$, $x = 2$ then y is:

- A. 2
- B. 3
- C. 4
- D. 5

6. Which ordered pair satisfy the equation $y = 2x$:

- A. (1, 2)
- B. (2, 1)
- C. (2, 2)
- D. (0, 1)

7. The real numbers x , y of the ordered pair (x, y) are called ____ of point $P(x,y)$ in a plane.

- A. co-ordinates
- B. x co-ordinates
- D. ordinate

8. Cartesian plane is divided into __ quadrants.

- A. Two
- B. Three
- C. Four
- D. Five

9. The point of intersection of two coordinate axes is called:

- A. Origin
- B. Centre
- C. X-coordinate
- D. y-coordinate

10. The x-coordinate of a point is called__

- A. Origin
- B. abscissa
- C. y-coordinate
- D. Ordinate

11. The y-coordinate of a point is called:

- A. Origin
- B. x-coordinate
- C. y-coordinate
- D. ordinate

12. The set of points which lie on the same line are called __ points.

- A. Collinear
- B. Similar
- C. Common
- D. None of these

13. The plane formed by two straight lines perpendicular to each other is called:

- A. Cartesian plane
- B. Coordinate axes
- C. Plane
- D. None of these

14. An ordered pair is a pair of elements in which elements are written in specific:

- A. Order
- B. Array
- C. Point
- D. None

15. Point () 1,2 - lies in quadrant.

- A. I
- B. II
- C. III
- D. IV

16. Point () 1,1 lies in quadrant.

- A. I
- B. II
- C. III
- D. IV

17. Point () 1, 3 - lies in quadrant.

- A. I
- B. II
- C. III
- D. IV

18. Which of the following points is on the origin?

- A. () 0,0
- B. () 2, 3 - -
- C. () 0,2
- D. () 4,0

19. Which of the following lines is parallel to x-axis?

- A. $0 x =$
- B. $3 x = -$
- C. $3 x =$
- D. $3 y = -$

20. Which of the following lines is parallel to y-axis?

- A. $2 y x =$
- B. $3 x = -$
- C. $3 y =$
- D. $4 1 y x = +$

Chapter 9

1. Distance between points (0, 0) and (1, 1) is:

- A. 0
- B. 1
- C. 2
- D. 2

2. Distance between the points (1, 0) and (0, 1) is:

- A. 0
- B. 1
- C. 2
- D. 2

3. Mid-point of the points (2, 2) and (0,0) is:

- A. (1, 1)
- B. (1, 0)
- C. (0, 1)
- D. (-1, -1)

4. Mid-point of the points (2, -2) and (-2, 2) is:

- A. (2, 2)
- B. (-2, -2)
- C. (0, 0)
- D. (1, 1)

5. A triangle having all sides equal is called:

- A. Isosceles
- B. Scalene
- C. Equilateral
- D. None of these

6. A triangle having all sides different is called:

- A. Isosceles
- B. Scalene
- C. Equilateral
- D. None of these

7. The points P, Q and R are collinear if:

- A. $PQ + QR = PR$
- B. $PQ + QR > PR$
- C. $PQ + QR < PR$
- D. None of these

8. The distance between two points P(x1, y1) and Q (x2, y2) in the coordinate plane is: $d > 0$

- A. $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- B. $d = \sqrt{(x_2 + x_1)^2 + (y_2 + y_1)^2}$
- C. $d = \sqrt{(x_2 - x_1)^2 + (y_2 + y_1)^2}$
- D. $d = \sqrt{(x_2 + x_1)^2 + (y_2 - y_1)^2}$

9. A triangle having two sides equal is called:

- A. Isosceles
- B. Scalene
- C. Equilateral
- D. None of these

10. A right angled triangle is that in which one of the angles has measure equal to:

- A. 80°
- B. 90°
- C. 45°
- D. 60°

11. In a right angled triangle ABC, where $\angle ACB = 90^\circ$.

- A. $AB^2 = BC^2 + CA^2$
- B. $AB^2 = BC^2 - CA^2$
- C. $AB^2 = BC^2 + CA^2$
- D. $AB^2 = BC^2 - CA^2$

12. In a ΔABC , if $AB = BC = CA$, the triangle will be:

- A. isosceles
- B. scalene
- C. equilateral
- D. right-angled

13. If three or more than three points lie on the same line then points are called ____.

- A. non-collinear
- B. collinear
- C. parallel
- D. perpendicular

14. A _____ has two end points. 09309076

- A. line
- B. line segment
- C. ray
- D. triangle

15. A line segment has __ midpoint. 09309077

- A. one
- B. two
- C. three
- D. four

16. Each side of triangle has ____ collinear vertices. 09309078

- A. one
- B. two
- C. three
- D. four

Chapter 10 - Congruent Triangle

1. _____ triangle is an equiangular triangle.

- A. A scalene
- B. An isosceles
- C. An equilateral
- D. A right angled

2. A _____ has two end points

- A. line
- B. line segment
- C. ray
- D. angle

3. Three points are said to be collinear, if they lie on the same:

- A. plane
- B. line
- C. interior
- D. area

4. Two lines can intersect at:

- A. one point
- B. two points
- C. no point
- D. infinite point

5. Two _____ lines cannot intersect each other:

- A. perpendicular
- B. parallel
- C. non-parallel
- D. coplanar

6. All the medians of _____ triangle are equal in measure.

- A. a scalene
- B. an isosceles
- C. an equilateral
- D. a right angled

7. If two angles of a triangle are congruent then the sides opposite to them are

- A. congruent
- B. equal
- C. non congruent
- D. similar

8. Symbol for congruent is:

- A. $\square\square\rightarrow$
- B. N
- C. $\square$
- D. =

9. Symbol for correspondence is

- A. $\square\square\rightarrow$
- B. N
- C. $\square$
- D. =

10. How many end points has a ray?

- A. 1
- B. 2
- C. 3
- D. 4

11. Symbolically two congruent triangles ABC and PQR are written as:

- A. $ABC \ PQR \ \square = \square$
- B. $ABC \ PQR \ \square \square$
- C. $ABC \ PQR \ \square \square\square$
- D. $ABC \ PQR \ \square \neq\square$

12. Which of the following is postulate?

- A. . . . S S S S S $\square$
- B. . . . S A A S A A $\square$
- C. . . H S H S $\square$
- D. . . S A S

13. If sum of measures of two angles is 180o then angles are ____ angles.

- A. Complementary
- B. Supplementary
- C. Equal
- D. Right

14. If sum of measure of two angles is 90o then angles are ____ angles.

- A. Complementary
- B. Supplementary
- C. Congruent
- D. Acute

15. Hypotenuse is a side opposite to ____ in right angled triangle.

- A. 30o
- B. 60o
- C. 90o
- D. 120o

16. In equilateral triangle each angle is of ____.

- A. 30o
- B. 60o
- C. 90o
- D. 180o

17. Corresponding sides of congruent triangles are:

- A. equal
- B. different
- C. perpendicular
- D. parallel

18. Median bisecting the base angle of an isosceles triangle bisects the ____ angle.

- A. base
- B. vertical
- C. right
- D. acute

19. The median bisecting the base of an isosceles triangle is ____ to the base.

- A. parallel
- B. perpendicular
- C. collinear
- D. adjacent

20. Corresponding angles of congruent triangles are:

- A. congruent
- B. non-congruent
- C. unequal
- D. supplementary

21. Any two medians of an ____ triangle equal in measure.

- A. isosceles
- B. equilateral
- C. acute
- D. obtuse

22. Sum of all the interior angles of a triangle is. a) 90° b) 150° c) 180° d) 360°

Choose the correct answer.

Chapter 11

1. In a parallelogram opposite sides are...

- A. different
- B. perpendicular
- C. congruent
- D. intersecting

2. In a parallelogram opposite angles are

- A. parallel
- B. congruent
- C. complementary
- D. adjacent

3. Diagonals of a parallelogram each other at a point

- A. perpendicular to
- B. intersect
- C. equal to
- D. parallel to

4. Medians of triangle are.....

- A. equal
- B. concurrent
- C. congruent
- D. parallel

5. Diagonal of a parallelogram divides the parallelogram into triangles.

- A. two equal
- B. two different
- C. three different
- D. three equal

6. In a parallelogram shown in fig. $y^\circ = \dots\dots$

- A. 115°
- B. 90°
- C. 75°
- D. 105°

7. In a parallelogram shown in fig. $x^\circ = \dots\dots$

- A. 115°
- B. 90°
- C. 75°
- D. 105°

8. In a parallelogram shown in fig. $x^\circ \dots\dots\dots$

- A. 55°
- B. 5°
- C. 44°
- D. 125°

9. In a parallelogram shown in fig. $m = \dots\dots\dots$

- A. 8
- B. 10
- C. 2
- D. 4

10. In $\square ABC$ E and D are midpoints of the sides AB and AC respectively. Find the value of $m \angle DE$.

- A. 6°
- B. 9°
- C. 18°
- D. 10°

11. In parallelogram congruent parts are:

- A. Opposite sides
- B. Diagonals
- C. Opposite angles
- D. Opposite sides and angles

12. Alternate angles on parallel lines intersected by a transversal are ____.

- A. Congruent
- B. Non-congruent
- C. Complementary
- D. Supplementary

PARALLELOGRAMS AND TRIANGLES

Chapter 12 - Line Bisector And Angle Bisectors

1. Bisection means to divide into ____ equal parts

- A. Two
- B. Three
- C. Four
- D. Five

2. ____ of line segment means to draw perpendicular which passes through the midpoint of line segment.

- A. Right bisection
- B. Bisection
- C. Congruent
- D. Mid-point

3. Any point on the ____ of a line segment is equidistant from its end points:

- A. Right bisector
- B. Median
- D. Altitude

- 4. Any point equidistant from the end points of line segment is on the ____ of it:**
- A. Right bisector
 - B. Median
 - D. Altitude
- 5. The bisectors of the angles of a triangle are:**
- A. Concurrent
 - B. Congruent
 - C. Parallel
 - D. None
- 6. Bisection of an angle means to draw a ray to divide the given angle into ____ equal parts:**
- A. Four
 - B. Three
 - C. Two
 - D. Five
- 7. If $\overleftrightarrow{CD}$ is right bisector of line segment AB then: (i) $m\angle O A =$**
- A. $m\angle O Q$
 - B. $m\angle O B$
 - C. $m\angle A Q$
 - D. $m\angle B Q$
- 8. If $\overleftrightarrow{CD}$ is right bisector of line segment AB , then $m\angle A Q =$ ____**
- A. $m\angle O A$
 - B. $m\angle O B$
 - C. $m\angle B Q$
 - D. $m\angle O D$
- 9. The right bisectors of the sides of an acute triangle intersect each other ____ the triangle.**
- A. Inside
 - B. Outside
 - C. Midpoint
 - D. None
- 10. The right bisectors of the sides of a right triangle intersect each other on the ____**
- A. Vertex
 - B. Midpoint
 - C. Hypotenuse
 - D. None
- 11. The right bisectors of the sides of an obtuse triangle intersect each other ____ the triangle.**
- A. Outside
 - B. Inside
 - C. Midpoint
 - D. None
- 12. The point of line segment through which the right bisector passes is called its ____ point.**
- A. end
 - B. mid
 - C. non-collinear
 - D. trisection
- 13. The point of intersection of right bisectors of sides of a triangle is equidistant from the ____ of triangle.**
- A. sides
 - B. vertices
 - C. centre
 - D. angles
- 14. The altitudes of a triangle are**

Chapter 13 - Sides And Angles Of A Triangle

1. Which of the following sets of lengths can be the lengths of the sides of a triangle:

- A. 2cm, 3cm, 5cm
- B. 3cm, 4cm, 5cm
- C. 2cm, 4cm, 7cm
- D. 1cm, 2cm, 3cm

2. Two sides of a triangle measure 10cm and 15cm. Which of the following measure is possible for the third side

- A. 5cm
- B. 20cm
- C. 25cm
- D. 30cm

3. The angle opposite to the longer side is:

- A. Greater
- B. Shorter
- C. Equal
- D. None

4. In right angle triangle greater angle of:

- A. 60o
- B. 30o
- C. 75o
- D. 90o

5. In an isosceles right-angled triangle angles other than right angle are each of:

- A. 40o
- B. 45o
- C. 50o
- D. 55o

6. A triangle having two congruent sides is called ___ triangle.

- A. Equilateral
- B. Isosceles
- C. Right
- D. None

7. Perpendicular to line form an angle of ___

- A. 30o
- B. 60o
- C. 90o
- D. 120o

8. Sum of two sides of triangle is ___ than the third.

- A. Greater
- B. Smaller
- C. Equal
- D. None

9. The distance between a line and a point on it is ___

- A. Zero
- B. One
- C. Equal
- D. None

10. The difference of two sides of a triangle is ___ the third side.

- A. greater than
- B. smaller than
- C. equal to
- D. congruent to

11. In a triangle, the side opposite to greater angle is ____.

- A. smaller
- B. greater
- C. equal
- D. congruent

12. In a triangle the angles opposite to congruent sides are ____.

- A. congruent
- B. concurrent
- C. unequal
- D. non- congruent

13. In a triangle, the side opposite to smaller angle is ____.

- A. smaller
- B. greater
- C. congruent
- D. concurrent

14. An exterior angle of a triangle is ____ non-adjacent interior angle.

- A. equal to
- B. smaller than
- C. greater than
- D. congruent to

15. For a $\triangle ABC$, which of the following is true?

- A. $m\angle A + m\angle B + m\angle C < 180^\circ$
- B. $m\angle A + m\angle B + m\angle C > 180^\circ$
- C. $m\angle A + m\angle B + m\angle C = 180^\circ$
- D. $m\angle A + m\angle B + m\angle C = 360^\circ$

16. What is the supplement of a right angle?

- A. 60°
- B. 90°
- C. 120°
- D. 180°

17. The sum of the measures of two sides of a triangle is greater than ____ the measure of the median which bisects the third side.

- A. twice
- B. thrice
- C. hypotenuse
- D. angles

18. In an obtuse angled triangle, the side opposite to the obtuse angle is

Chapter 14 - Ratio And Proportion

1. One and only one line can be drawn through ____ points.

- A. Two
- B. Three
- C. Four
- D. Five

2. The ratio between two alike quantities is defined as:

- A. $a : b$
- B. $b - a$
- C. $a : b = c : d$
- D. $a + b$

3. If a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the __ side.

- A. Third
- B. Fourth
- C. Second
- D. None

4. Two triangles are said to be similar if these are equiangular and their corresponding sides are

- A. Proportional
- B. congruent
- C. concurrent
- D. None

5. In $\triangle LMN$ shown in the figure $MN \parallel PQ$ if $mLM = 5\text{cm}$, $mLP = 2.5\text{cm}$, $mLQ = 2.3\text{cm}$ then __ $mLN =$:

- A.

6. A line segment has _____ mid-point

- A. only one
- B. only two
- C. only three
- D. infinite

7. Ratio has no

- A. value
- B. symbol
- C. unit
- D. importance

8. Statement of equality of two ratios is called

- A. double ratio
- B. simple ratios
- C. proportion
- D. Relation

9. The symbol used for similarity is.....

- A. =
- B. $\square$
- C.
- D. $\square$

10. The symbol used for congruency is

- A. =
- B. $\square$
- C.
- D. $\square$

11. The symbol used for ratio is

- A.
- B. $\square$
- C. $\sim$
- D.

12. The ratio between two alike quantities has no.....

- A. value
- B. symbol
- C. unit
- D. importance

13. The symbol used for line AB is

- A. AB
- B. AB
- C. AB $\square \rightarrow$
- D. AB $\square \rightarrow$

14. The symbol used for ray AB is

- A. AB
- B. $\overleftrightarrow{AB}$
- C. $\overrightarrow{AB}$
- D. $\overleftarrow{AB}$

15. The symbol used for line segment AB is

- A. AB
- B. $\overleftrightarrow{AB}$
- C. $\overrightarrow{AB}$
- D. $\overleftarrow{AB}$

16. $\overrightarrow{AB}$ stands for

- A. line AB
- B. Ray AB
- C. line segment AB
- D. points AB

17. Proportion is a equality of ratios.

- A. Two
- B. Three
- C. Four
- D. Five

18. Similar triangles are of the same shape but in sizes.

- A. The same
- B. Different
- C. Both
- D. None of these

19. $\perp$ is the symbol of:

- A. equal
- B. parallel
- C. perpendicular
- D. congruent

Chapter 15

1. In a right angled triangle, the square of the length of hypotenuse is equal to the ____ of the squares of the lengths of the other two sides.

- A. Sum
- B. Difference
- C. Zero
- D. None of these

2. If the square of one side of a triangle is equal to the sum of the squares of the other two sides then the triangle is a ____ triangle.

- A. Right angled
- B. Acute angled
- C. Obtuse angled
- D. None of these

3. Let c be the longest of the sides a, b and c of a triangle. If $a^2 + b^2 = c^2$, then the triangle is ____:

- A. Right
- B. Acute
- C. Obtuse
- D. None of these

4. Let c be the longest of the sides a, b and c of a triangle. If $a^2 + b^2 > c^2$ then triangle is:

- A. Acute
- B. Right
- C. Obtuse
- D. None of these

- 5. Let c be the longest of the sides a , b and c of a triangle. If $a^2 + b^2 < c^2$, then the triangle is:**
- A. Acute
 - B. Right
 - C. Obtuse
 - D. None of these
- 6. If 3cm and 4cm are two sides of a right angled triangle, then hypotenuse is;**
- A. 5cm
 - B. 3cm
 - C. 4cm
 - D. 2cm
- 7. In right triangle ____ is a side opposite to right angle.**
- A. Base
 - B. Perpendicular
 - C. Hypotenuse
 - D. None
- 8. In the fig.**
- A. $x = 6\text{cm}$
 - B. $x = 8\text{cm}$
 - C. $x = 10\text{cm}$
 - D. $x = 16\text{cm}$
- 9. In the fig.**
- A. $x = 5\text{cm}$
 - B. $x = 8\text{cm}$
 - C. $x = 12\text{cm}$
 - D. $x = 18\text{cm}$
- 10. In the fig.**
- A. $x = 2\text{cm}$
 - B. $x = 1\text{cm}$
 - C. $x = 2\text{cm}$
 - D. $x = 3\text{cm}$
- 11. In right angled triangle greater angle is ____.**
- A. 30°
 - B. 60°
 - C. 90°
 - D. 120°
- 12. In right angled triangle one angle is 90° and other two angles are ____**
- A. obtuse
 - B. acute
 - C. right
 - D. supplementary
- 13. If hypotenuse of an isosceles right angled triangle is 2 then each of other side is:**
- A. 1cm
 - B. 2cm
 - C. 3cm
 - D. 4cm
- 14. In right angled triangle which side is the longest side?**
- A. perpendicular
 - B. base
 - C. hypotenuse
 - D. none of these

15. In right angled triangle if 90° m B $\square$ = then which of the following is true?

- A. $2 \angle 2 \angle a \angle b \angle c + =$
- B. $2 \angle 2 \angle a \angle c \angle b + =$
- C. $2 \angle 2 \angle b \angle c \angle a + =$
- D. $2 \angle 2 \angle a \angle c \angle b - =$

16. In a Isosceles right angled triangle two acute angles are equal to:

- A. 30°
 - B. 45°
 - C. 60°
 - D. 90°
- 6cm 10cm x x 13cm 5cm x 1cm 2 cm

Chapter 16

1. The region enclosed by the bounding lines of a closed figure is called the __ of the figure:

- A. Area
- B. Circle
- C. Boundary
- D. None

2. Base $\times$ altitude =

- A. Area of parallelogram
- B. Area of square
- C. Area of Rectangular
- D. Area of Triangle

3. The union of a rectangle and its interior is called:

- A. Circle region
- B. Rectangular region
- C. Triangle region
- D. None

4. If a is the side of a square, its area will be equal to...

- A. a square unit
- B. a^2 square units
- C. a^3 square units
- D. a^4 square units

5. The union of a triangle and its interior is called as:

- A. Triangular region
- B. Rectangular region
- C. Circle region
- D. None of these

6. Altitude of a triangle means perpendicular distance to base from its opposite__

- A. Vertex
- B. Side
- C. Midpoint
- D. None

7. Area of given figure is.....

- A. 18cm
- B. 9cm
- C. 18cm^2
- D. 9cm^2

8. Area of given figure is.....

- A. 4cm
- B. 8cm^2
- C. 16cm
- D. 16cm^2

9. Area of given figure is.....

- A. 4cm²
- B. 12cm²
- C. 32cm
- D. 32cm²

10. Area of given figure is....

- A. 160cm²
- B. 80cm²
- C. 80cm
- D. 160cm

11. Area of triangle is

- A. $A = \frac{1}{2} \text{ Base} \times \text{Height}$
- B. $A = \text{Base} \times \text{Height}$
- C. $A = L \times w$
- D. $A = L^2$

12. Area of square is

- A. $A = \frac{1}{2} \text{ Base} \times \text{Height}$
- B. $A = \text{Base} \times \text{Height}$
- C. $A = L \times w$
- D. $A = L^2$

13. Area of rectangle is

- A. $A = \frac{1}{2} \text{ Base} \times \text{Height}$
- B. $A = \text{Base} \times \text{Height}$
- C. $A = L \times w$
- D. $A = L^2$

14. Area of parallelogram is ...

- A. $A = \frac{1}{2} \text{ Base} \times \text{Height}$
- B. $A = \text{Base} \times \text{Height}$
- C. $A = L \times w$
- D. $A = L^2$

15. If the length and breadth of a rectangle are 'a' and 'b' then its area will be:

- A. $a + b$
- B. $a \times b$
- C. $a b -$
- D. $a = b$

16. In most cases similar figures have ____ areas.

- A. same
- B. different
- C. equal
- D. congruent

17. All congruent figures have ____ areas.

- A. same
- B. different
- C. zero
- D. non-congruent

18. Area of a geometrical figure is always

Chapter 17

1. A triangle having two sides congruent is called: ____

- A. Scalene
- B. Right angled
- C. Equilateral
- D. Isosceles

- 2. A quadrilateral having each angle equal to 90° is called ____**
- A. Parallelogram
 - B. Rectangle
 - C. Trapezium
 - D. Rhombus
- 3. The right bisectors of the three sides of a triangle are ____**
- A. Congruent
 - B. Collinear
 - C. Concurrent
 - D. Parallel
- 4. The ____ altitudes of an isosceles triangle are congruent:**
- A. Two
 - B. Three
 - C. Four
 - D. None
- 5. A point equidistant from the end points of a line segment is on its ____**
- A. Bisector
 - B. Right bisector
 - C. Perpendicular
 - D. Median
- 6. ____ congruent triangles can be made by joining the mid points of the sides of a triangle:**
- A. Three
 - B. Four
 - C. Five
 - D. Two
- 7. The diagonals of a parallelogram ____ each other:**
- A. Bisect
 - B. Trisect
 - C. Bisect at right angle
 - D. None of these
- 8. The medians of a triangle cut each other in the ratio:**
- A. 4:1
 - B. 3:1
 - C. 2:1
 - D. 1:1
- 9. One angle on the base of an isosceles triangle is 30° . What is the measure of its vertical angle:**
- A. 30°
 - B. 60°
 - C. 90°
 - D. 120°
- 10. If the three altitudes of a triangle are congruent then the triangle is _**
- A. Equilateral
 - B. Right angled
 - C. Isosceles
 - D. Acute angled
- 11. If two medians of a triangle are congruent then the triangle will be:**
- A. Isosceles
 - B. Equilateral
 - C. Right angled
 - D. Acute angled

- 12. A line segment joining a vertex of a triangle to the midpoint of its opposite side is called a ___ of the triangle:**
- A. Altitude
 - B. Median
 - C. Angle bisector
 - D. Right bisector
- 13. A line segment from a vertex of triangle perpendicular to the line containing the opposite side, is called an ___ of the triangle:**
- A. Altitude
 - B. Median
 - C. Angle bisector
 - D. Right bisector
- 14. The point of concurrency of the three altitudes of a $\triangle$ is called its ___**
- A. Ortho centre
 - B. In centre
 - C. Circumcentre
 - D. None
- 15. The internal bisectors of the angles of a triangle meet at a point called the ____ of the triangle:**
- A. In centre
 - B. Ortho centre
 - C. Circumcentre
- 16. The point of concurrency of the three perpendicular bisectors of the sides of a triangle is called the ___ of the triangle.**
- A. Circumcentre
 - B. In centre
 - C. Ortho centre
 - D. None
- 17. Point of concurrency of three medians of a triangle is called.**
- A. In centre three
 - B. Ortho centre
 - C. Centroid
 - D. Circumcentre
- 18. Sum of interior angles of a triangle is**
- A. 60o
 - B. 120o
 - C. 180o
 - D. 240o
- 19. The side opposite to right angle in right angled triangle is called....**
- A. Base
 - B. Perpendicular
 - C. Hypotenuse
 - D. Altitude
- 20. The altitudes of a right angled triangle are concurrent at the**
- A. Midpoint of hypotenuse
 - B. Vertex of right angle
 - C. Midpoint of base
 - D. Vertical angle
- 21. The triangles are said to be if they are equiangular.**
- A. Congruent
 - B. Similar
 - C. Equal
 - D. Scalene

22. All the right bisectors of sides of triangle are concurrent.

- A. One
- B. Two
- C. Three
- D. Four

23. All the three bisectors of angles of triangle are.....

- A. Congruent
- B. Concurrent
- C. Parallel
- D. Perpendicular

24. All the three medians of a triangle are.....

- A. Congruent
- B. Concurrent
- C. Parallel
- D. Perpendicular

25. All the three altitudes of a triangle are.....

- A. Congruent
- B. Concurrent
- C. Parallel
- D. Perpendicular

26. In-centre is the point of concurrency of three..... of triangle.

- A. Right bisectors
- B. Angle bisectors
- C. Altitudes
- D. Medians

27. Circumcentre is point of concurrency of three of three..... of triangle.

- A. right bisectors
- B. angle bisectors
- C. altitudes
- D. medians

28. Centroid is the point of concurrency of three..... of triangle.

- A. right bisectors
- B. angle bisectors
- C. altitudes
- D. medians

29. Three or more than three lines passing through the same point are called Lines.

- A. congruent
- B. concurrent
- C. parallel
- D. perpendicular

30. The common point of three or more than three lines is called.....

- A. central point
- B. point of concurrency
- C. vertex
- D. centroid

31. In right-angled triangle if one angle is 30° , then other angle will be.....:

- A. 15°
- B. 30°
- C. 45°
- D. 60°

32. In right-angled triangle if one angle is 60° , then other angle will be.....:

- A. 15°
- B. 30°
- C. 45°
- D. 6°

Answer Key

Answers are placed chapter-wise for quick checking.

Chapter 1

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	C	3	A	4	C	5	B
6	A	7	A	8	C	9	D	10	A
11	A	12	A	13	A	14	B	15	C
16	B	17	A	18	B	19	D	20	A

Chapter 2

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	C	3	A	4	-	5	B
6	C	7	C	8	D	9	B	10	A
11	A	12	C	13	B	14	A	15	C
16	C	17	C	18	A	19	A	20	A
21	A	22	A	23	B	24	A	25	A
26	A	27	C	28	A	29	C	30	B

Chapter 3

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	D	4	A	5	-
6	A	7	D	8	C	9	A	10	C
11	B	12	B	13	A	14	B	15	A
16	A	17	A	18	A	19	D	20	D
21	D	22	A	23	A	24	B		

Chapter 4

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	D	3	B	4	A	5	B
6	B	7	-	8	C	9	A	10	A
11	A	12	A	13	B	14	A	15	B
16	B	17	C	18	C	19	D	20	A
21	D								

Chapter 5 - Factorization

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	C	3	D	4	B	5	C
6	C	7	C	8	-	9	A	10	A
11	A	12	B	13	B	14	A	15	A

16	C	17	D	18	B	19	D		
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Chapter 6

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	A	3	C	4	B	5	A
6	A	7	A	8	B	9	C	10	C
11	C	12	A	13	A	14	D	15	B
16	C	17	B	18	A	19	A	20	A
21	A	22	B	23	B	24	C	25	A
26	-	27	-	28	-				

Chapter 7

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	C	3	C	4	B	5	C
6	D	7	A	8	A	9	A	10	A
11	A	12	A	13	B	14	A	15	A
16	A	17	A	18	A	19	A	20	A
21	C	22	C	23	A	24	B	25	C
26	A	27	A						

Chapter 8

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	C	3	D	4	C	5	D
6	A	7	A	8	C	9	A	10	B
11	D	12	A	13	A	14	A	15	B
16	A	17	D	18	C	19	A	20	A

Chapter 9

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	C	3	A	4	C	5	C
6	B	7	A	8	A	9	A	10	B
11	A	12	C	13	B	14	B	15	A
16	B								

Chapter 10 - Congruent Triangle

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	B	4	A	5	B
6	C	7	A	8	C	9	A	10	A
11	C	12	D	13	B	14	A	15	-
16	B	17	A	18	B	19	B	20	A

21	B	22	C						
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Chapter 11

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	B	4	B	5	A
6	C	7	D	8	B	9	C	10	B
11	-	12	-						

Chapter 12 - Line Bisector And Angle Bisectors

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	A	3	A	4	A	5	A
6	C	7	B	8	C	9	A	10	C
11	A	12	B	13	C	14	B		

Chapter 13 - Sides And Angles Of A Triangle

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	A	4	D	5	B
6	B	7	C	8	A	9	A	10	B
11	B	12	A	13	A	14	C	15	C
16	B	17	A	18	B				

Chapter 14 - Ratio And Proportion

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	A	3	A	4	A	5	A
6	A	7	C	8	C	9	C	10	B
11	D	12	C	13	C	14	D	15	B
16	A	17	A	18	B	19	C		

Chapter 15

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	A	3	A	4	A	5	C
6	A	7	C	8	B	9	C	10	B
11	C	12	B	13	A	14	C	15	B
16	B								

Chapter 16

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	A	3	B	4	B	5	A
6	A	7	C	8	D	9	D	10	B
11	A	12	D	13	C	14	B	15	B

16	B	17	A	18	B				
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Chapter 17

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	2	B	3	C	4	A	5	B
6	B	7	A	8	C	9	D	10	A
11	A	12	B	13	A	14	A	15	A
16	A	17	C	18	C	19	C	20	B
21	B	22	C	23	B	24	B	25	B
26	B	27	A	28	D	29	B	30	B
31	D	32	B						